DealScout Viva Companion

Use this guide to explain the system in simple, accurate language. Start with the problem, show the workflow, then discuss technology and limitations.

- One-line definition: DealScout compares normalized product offers and highlights the lowest listed price.
- Demonstration order: search → comparison → best price → login → insights.
- Be honest that live mode needs a provider key.

VIVA TIP

Speak clearly, demonstrate the real flow, and separate implemented behavior from planned enhancements.

60-Second Introduction

DealScout reduces the effort of visiting many stores separately. A user enters a product name, chooses a category and source mode, and receives normalized offers.

- The backend validates the query.
- Offers use one common ProductOffer structure.
- The minimum numeric price becomes the best deal.
- Each search is stored for insights.

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Problem and Objectives

The main problem is inconsistent and scattered e-commerce information.

- Save comparison time.
- Improve clarity through a common table.
- Show seller, rating, discount, and delivery with price.
- Record searches for later review.
- Support portable setup.

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Architecture Answer

The system follows a client-server architecture.

- React and TypeScript run in the browser.
- Nginx serves the frontend and proxies /api.
- FastAPI handles validation and business logic.
- MongoDB stores search logs.
- An optional provider supplies live shopping results.

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Search Algorithm

Explain the algorithm step by step without jargon.

- Read and trim the query.
- Reject input shorter than two characters.
- Filter matching offers by query and category.
- Normalize every accepted source.
- Use minimum price to choose best_deal_id.
- Store the summary and return JSON.

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Database Answer

The main stored entity is SearchLog; ProductOffer is the normalized response structure.

- SearchLog: query, category, mode, count, best price, sources, timestamp.
- ProductOffer: product, source, seller, price, rating, delivery, discount.
- The recent-search query uses a descending timestamp index.

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Authentication Answer

The preview login creates an HTTP-only cookie session.

- The cookie travels automatically with same-origin requests.
- Browser JavaScript does not store a token.
- Analytics checks the session before returning data.
- Production would use persistent sessions and password hashing.

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API Questions

REST endpoints exchange JSON between the browser and server.

- GET /api/products/search — search and compare.
- POST /api/auth/login — create session.
- GET /api/auth/me — return current user or null.
- GET /api/analytics — protected search insights.
- HTTP 422 means request validation failed.

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Live vs Sample Data

This distinction is important during the viva.

- Sample catalog is local, stable, and always available.
- Live search requires SERPAPI_KEY on the backend.
- Without a key, preview results are explicitly labelled and not claimed as live.
- Provider keys never appear in frontend code.

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Testing Answer

Testing covers the backend contract, strict frontend types, and the browser journey.

- Positive API fields and status codes are asserted.
- A short invalid query proves validation.
- Typecheck catches interface mismatch.
- Browser test verifies search, login, insights, console, and network errors.

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Limitations and Future Scope

A strong viva answer states limitations before enhancements.

- Limited local catalog.
- No checkout or payment.
- Price may differ by tax, shipping, location, and variant.
- Preview login is not production identity.
- Future: verified providers, alerts, favorites, trends, and admin tools.

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Presentation Checklist

Prepare the environment before entering the viva.

- Open the app and keep the login ready.
- Search iPhone 15 in Sample prices.
- Show best-price logic and comparison rows.
- Sign in and open Insights.
- Keep Word/PDF report and source code available.
- Answer only what is implemented; call future work future work.

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